

1) Windows Sandbox: test scary stuff without risking your real PC

Windows Sandbox is one of the most underrated security features in Windows 11. It gives you a lightweight, isolated desktop environment where you can run apps or open files safely, and then throw the entire environment away when you're done. Anything you do inside it stays isolated from your main system.

Think of it like borrowing a clean, temporary laptop for five minutes—then returning it with amnesia.

When should you use Sandbox?

- You downloaded an installer from a sketchy link and your instincts are screaming.
- A client sent you an “urgent” ZIP file.
- You want to test a tool/plugin without polluting your system.
- You're troubleshooting whether an app crash is caused by your system or the app itself.

Requirements and install (the safe, official way)

Microsoft's install guidance boils down to:

- You must be on Windows 11 (supported editions)
- Virtualization must be enabled in firmware/BIOS
- Then you enable the Windows Sandbox feature.

Step-by-step: Enable Windows Sandbox

1. Open Start → search “Turn Windows features on or off”
2. In the list, find Windows Sandbox
3. Tick it → click OK
4. Restart when prompted

(Some newer Windows 11 builds also centralize virtualization toggles in a Settings page. If you see something like “Virtual Workspaces” in Settings, it may allow enabling Sandbox/Hyper-V from one place—but availability varies by build/edition.)

Step-by-step: Run Sandbox

1. Open Start → type Windows Sandbox → open
2. A clean Windows desktop appears—this is the sandboxed environment

The #1 Sandbox workflow: test a downloaded file safely

This is the “I don't trust this EXE” play.

1. On your main Windows, locate the file (e.g., in Downloads).
2. Open Sandbox.
3. Copy the file into the Sandbox window (copy/paste works).
4. Run it in Sandbox.
5. If it behaves suspiciously, close Sandbox. Done. No cleanup needed.

When you close Sandbox, the environment is discarded—Microsoft explicitly positions it as a disposable VM-like environment.

Bonus: Sandbox configuration files (.wsb) — make Sandbox behave how you want

Windows Sandbox supports simple configuration files with a .wsb extension that let you control things like mapped folders, networking, and startup commands.

Practical example: “Test Downloads safely” Sandbox

Microsoft even provides sample configs where you can map your Downloads folder read-only, disable networking, and open the folder automatically.

Fast Track rule: If you're testing an unknown file, consider disabling networking in the Sandbox config, because Sandbox networking is enabled by default and Microsoft warns it can expose your internal network if you run untrusted apps.

2) Hyper-V: full virtual machines (for devs, testers, power users)

Sandbox is a disposable scratchpad. Hyper-V is the real deal: full virtual machines with dedicated disk images, snapshots, networks, and long-term setups.

Who should use Hyper-V?

- Developers who need Windows/Linux VMs
- IT/admin users testing deployments
- Anyone running training labs, older software, or isolated environments
- People who need more control than Sandbox provides

Requirements (and why Home users hit a wall)

Microsoft's Hyper-V guidance states:

- Windows 11 Pro or Enterprise (Hyper-V role isn't available on Home)
- 64-bit CPU with SLAT
- Hardware virtualization support (VM Monitor Mode extensions)
- At least 4GB RAM (more is better)

Step-by-step: Enable Hyper-V (Windows 11 Pro/Enterprise)

Microsoft documents multiple ways; the cleanest is enabling the Windows optional feature.

Option A: PowerShell (fast)

1. Open PowerShell as Administrator
2. Run:
`Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All`
3. Restart when prompted

Option B: Windows Features UI

1. Start → search Turn Windows features on or off